Global mapping of soil organic carbon mean residence time reveals controls beyond climate

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Abstract

Keywords: soil organic carbon; mean residence time; random forest; accumulated local effects; Earth system models; soil biogeochemistry

1 Introduction

2 Methods

2.1 Conceptual framework

We treat soil organic carbon (SOC) mean residence time (MRT) as an emergent property of local ecological context rather than a function of climate alone. Under steady-state assumptions (constant SOC stock and input flux), MRT equals the ratio of the SOC stock to the annual carbon input flux and is the inverse of the apparent first-order decomposition rate constant k:

$$\mathrm{MRT} = \frac{\mathrm{SOC}_{\mathrm{stock}}}{C_{\mathrm{input}}} = \frac{1}{k} \tag{1}$$

To separate the contributions of different process classes to spatial variation in MRT, we designed a nested Random Forest (RF) modelling framework in which predictor variables are organised into four mechanistically distinct groups: Climate, Edaphic, Land Use, and Biological (Table 2). Seven model configurations (M1–M7) progressively combine these groups (Table 1), so that the change in cross-validated performance between successive models isolates the additional explanatory power of each process class while holding the climate baseline constant.

A key design constraint is the exclusion of productivity-related variables (net primary productivity, leaf area index) from the predictor set. Because MRT is calculated as SOC stock divided by an NPP-derived input flux (Eq. 1), including productivity as a predictor would introduce a circular dependency.

2.2 Soil data compilation

We compiled soil profile data from the World Soil Information Service (WoSIS; Batjes et al., 2024) point dataset (`sol_chem.pnts_horizons`), which integrates observations from multiple national and regional soil surveys worldwide. From the full dataset of horizon-level records, we retained only mineral soils by removing all records with organic

carbon (OC) $\geq$ 120 g kg^{-1} (the conventional threshold separating mineral from organic soil horizons). Records with missing OC were retained.

Horizon-level data were standardised to a 0–20 cm depth interval by depth-weighted averaging. For each unique soil profile (identified by site ID), we computed the overlap of each recorded horizon with the 0–20 cm target interval and calculated a thickness-weighted mean for seven soil variables: organic carbon, total nitrogen, pH (H_2O), clay content, silt content, sand content, and oven-dry bulk density. Site-level metadata (coordinates, observation date, source database) were carried forward as the most common value across horizons.

2.3 Bulk density gap-filling

Bulk density (BD) is required to convert OC concentration to SOC stock but is frequently unreported. We gap-filled missing BD values using a three-tier approach. First, a Random Forest model (`randomForest` R package; 200 trees) was trained on the subset of profiles with measured BD, using terrain metrics, Köppen zone, land cover fractions, SoilGrids soil properties (excluding BD), MODIS NPP, WRB functional group, and MODIS burn count as predictors. Ten-fold cross-validation confirmed that the RF model outperformed the SoilGrids BD product for sites with measured BD. Second, profiles lacking measured BD but with complete RF predictors received RF-predicted values. Third, remaining gaps were filled with SoilGrids BD as a fallback. Each profile was flagged with its BD source (measured, RF, or SoilGrids).

2.4 MRT calculation

SOC stock for the 0–20 cm layer was calculated as:

$$\mathrm{SOC_{stock}}\ [\mathrm{g\,C\,m^{-2}}] \;=\; \mathrm{OC} \;\times\; \mathrm{BD} \;\times\; 20 \;\times\; 10 \tag{2}$$

where OC is in g kg^{-1}, BD in g cm^{-3}, depth is 20 cm, and the factor 10 converts units.

Carbon input to soil was estimated as the belowground fraction of net primary produc-

tivity (BNPP). We used MODIS MOD17A3 annual NPP (2001–2020; Running and Zhao, 2021) extracted at each profile location, applying the MODIS scale factor ($\mathrm{NPP_{true}} = \mathrm{NPP_{raw}}/10$; g C $\mathrm{m^{-2}\ yr^{-1}}$). To maintain temporal consistency with the equilibrium assumption, only NPP values from years preceding each site's sampling date were used. Sites sampled before the MODIS era (pre-2001) or with fewer than three pre-sampling NPP years were excluded from the temporal statistics.

Belowground allocation fractions (BNPP/NPP) were assigned by dominant land cover class following Mokany et al. (2006), Malhi et al. (2011), and Litton and Giardina (2008): tree cover 0.35, shrubland 0.45, grassland 0.55, cropland 0.30, herbaceous wetland 0.50, mangroves 0.40, moss/lichen 0.60. Sites lacking land cover information received a global average of 0.40. Soil carbon input was then:

$$C_{\mathrm{input}}\ [\mathrm{g\,C\,m^{-2}\,yr^{-1}}] \;=\; \mathrm{NPP} \;\times\; f_{\mathrm{BNPP}} \tag{3}$$

and MRT was computed as the ratio of SOC stock to C_{input} (Eq. 1). Quality control flags were assigned: values with MRT $<$ 1 year or $>$ 1000 years, or with missing, negative, or infinite values were excluded. The upper and lower 1% tails of the remaining distribution were removed prior to model fitting.

2.5 Predictor variables

Predictor variables were organised into four groups reflecting distinct mechanisms influencing SOC turnover (Table 2).

2.5.1 Climate

Climate variables were derived from the ERA5-Land reanalysis monthly means (2001–2020; Muñoz-Sabater et al., 2021). From the original monthly fields of 2-m air temperature (`t2m`), soil temperature at layers 1 (0–7 cm; `stl1`) and 2 (7–28 cm; `stl2`), volumetric soil water at layers 1 (`swvl1`) and 2 (`swvl2`), snow cover fraction (`snowc`), and potential evaporation (`pev`), we calculated the following annual statistics: mean 2-m temperature, temperature seasonality (monthly standard deviation), depth-weighted mean soil tem-

perature for 0–20 cm (weights 0.35 for layer 1 and 0.65 for layer 2, proportional to the overlap with the 0–20 cm interval), depth-weighted mean soil moisture (same weights), mean snow cover fraction, and mean potential evaporation. Climate zone was represented by the Köppen-Geiger classification of Beck et al. (2023) at 30-class resolution (numeric codes 1–30), based on the 1991–2020 climatological period at ∼1 km native resolution, resampled to 0.1° by modal aggregation.

2.5.2 Edaphic

Soil properties were obtained from SoilGrids 250 m v2.0 (Poggio et al., 2021), depth-weighted to 0–20 cm: clay content, sand content, bulk density, coarse fragment volume, pH (H_2O), cation exchange capacity, and total nitrogen. All values were converted from SoilGrids integer encoding to standard units using published scale factors.

Terrain metrics were derived from the Copernicus GLO-30 Digital Elevation Model (European Space Agency and Airbus, 2022), aggregated to 0.1°: mean elevation, mean slope, northness and eastness (sine and cosine of aspect), and terrain ruggedness (focal standard deviation of elevation in a 3×3 cell window, ∼30 km).

Soil classification was based on the SoilGrids WRB Most Probable class (250 m; Poggio et al., 2021), aggregated to 0.1° by mode and reclassified into eight mechanistically defined functional groups: sandy/coarse (Arenosols, Podzols, Regosols), temperate agricultural (Luvisols, Cambisols, Phaeozems, Chernozems, Kastanozems, Albeluvisols), high-clay tropical (Acrisols, Ferralsols, Lixisols, Plinthosols, Nitisols, Alisols, Vertisols), arid/chemical (Calcisols, Gypsisols, Solonchaks, Solonetz, Durisols), poorly drained (Gleysols, Histosols, Fluvisols, Stagnosols, Planosols), volcanic (Andosols), shallow/young (Leptosols, Umbrisols), and cold/permafrost (Cryosols). These groups were encoded as integer codes for the RF models.

2.5.3 Land use and disturbance

Current land cover was characterised using ESA WorldCover 2021 (Zanaga et al., 2022), downloaded via the `geodata` R package. Eleven class-specific fractional cover layers (trees,

grassland, shrubland, cropland, built-up, bare, snow/ice, water, wetland, mangroves, moss/lichen) were aggregated from native ∼10 m to 0.1° by mean. Six of these fractions (trees, grassland, shrubland, cropland, bare, wetland) were included as continuous predictors.

Disturbance history was captured by two date-aware products. Forest loss year was extracted from the Hansen Global Forest Change v1.10 dataset (Hansen et al., 2013); a binary flag indicated whether loss occurred before each site's sampling date. Fire history was derived from the MODIS MCD64A1 burned area product (2001–2020; Giglio et al., 2018), with cumulative burn count before sampling used as a continuous predictor.

2.5.4 Biological

Microbial community structure was represented by three independent, spatially explicit datasets. Global soil fungal proportion (fungal biomass as a fraction of total microbial biomass) was obtained from Yu et al. (2022) at ∼1 km resolution. Root colonisation rates by arbuscular mycorrhizal (AM) and ectomycorrhizal (EcM) fungi were taken from the global maps of Barceló et al. (2023), along with a derived EcM:AM root colonisation ratio. Mycorrhizal richness and endemism (rarity-weighted richness) for both AM and EcM fungi were sourced from the SPUN Global Underground Atlas (Van Nuland et al., 2025), together with a derived EcM:AM richness ratio. All biological layers were aggregated to 0.1° by bilinear resampling.

2.5.5 Spatial alignment

All raster covariates were resampled to a common 0.1° (∼11 km) global grid in WGS84 (EPSG:4326). Continuous variables were resampled using bilinear interpolation; categorical variables (Köppen zone, WRB class) used modal aggregation. Point-level extraction was performed at native resolution where available and from the 0.1° layers otherwise.

2.6 Random Forest modelling

All RF models were fitted using the `ranger` R package (Wright and Ziegler, 2017) with 500 trees, a minimum node size of 5, and the default m_{try} ($= \lfloor \sqrt{p} \rfloor$, where p is the number of predictors). The response variable was log-transformed MRT (ln MRT); predictions were back-transformed via exponentiation. Parallel computation used all available cores minus one.

Seven nested model configurations were fitted (Table 1): M1 (Climate only), M2 (Climate + Edaphic), M3 (Climate + Land Use), M4 (Climate + Biological), M5 (Climate + Edaphic + Land Use), M6 (Climate + Edaphic + Biological), and M7 (Full: all four groups). This design holds the climate baseline constant across all models, so that the change in performance between M1 and each subsequent model directly quantifies the additional variance explained by each process class.

Model accuracy was assessed by spatial block cross-validation to avoid spatial autocorrelation-driven overfitting. The global domain was divided into $2^\circ \times 2^\circ$ grid cells ($\sim$200 km at the equator); grid cells were randomly assigned to one of ten folds (stratified by cell, not by individual observation). For each fold, the model was trained on the remaining nine folds and predictions were collected for the held-out block. Cross-validated R^2, RMSE, and MAE were computed on the log-scale predictions pooled across all folds. Out-of-bag (OOB) statistics from a final model fitted on all data served as an additional reference. Permutation-based variable importance was extracted from the final M7 model.

2.7 Global prediction

Each of the seven trained models was applied to the aligned 0.1° global raster stack to produce a spatially continuous MRT prediction map. Predictions were generated in log space and back-transformed to years. Difference maps between successive model configurations (e.g., M2 − M1, M7 − M1) were computed to visualise the spatial signature of each process class on predicted MRT.

2.8 Sensitivity analysis

We used Accumulated Local Effects (ALE; Apley and Zhu, 2020) rather than partial dependence plots to characterise the marginal response of MRT to each continuous predictor. ALE avoids the extrapolation artefacts inherent in partial dependence by evaluating prediction differences only within narrow quantile-based bins of the focal variable, so that correlated predictors retain their naturally co-occurring values.

The ALE analysis was conducted on a spatially stratified random subsample of 100 000 pixels drawn from the global 0.1° prediction grid (stratified by 5° latitude–longitude cells to ensure geographic representativeness). Each continuous predictor was divided into 40 quantile-based bins. Within each bin, the focal variable was nudged to the bin's upper and lower edge while all other variables were held at their observed values; the local prediction difference was accumulated across bins and centred to zero at the distribution's lower edge to produce a response curve in native MRT units (years). Categorical predictors (Köppen zone, WRB functional group code) were excluded from ALE.

To examine latitudinal variation in predictor responses, ALE curves were additionally computed within four latitude bands: southern extratropics ($<-23.5°$), tropics ($-23.5°$ to $23.5°$), northern temperate ($23.5°$ to $66.5°$), and boreal/Arctic ($>66.5°$).

2.9 ESM benchmarking

[To be completed once CMIP comparison analysis is finalised.]

2.10 Software and reproducibility

All analyses were conducted in R ($\geq$ 4.3) using the following key packages: `terra` (Hijmans, 2024) for raster and vector operations, `sf` for spatial data, `ranger` (Wright and Ziegler, 2017) for Random Forest modelling, `randomForest` for bulk density gap-filling, `geodata` for ESA WorldCover download, `ggplot2` for visualisation, and `dplyr/tidyr` for data manipulation. ERA5-Land data were downloaded from the Copernicus Climate Data Store; SoilGrids layers were accessed via ISRIC's cloud-optimised GeoTIFF service. The analysis pipeline consists of 18 numbered R scripts executing sequentially from data

loading through global prediction and sensitivity analysis. Code and intermediate outputs are archived at [DOI].

3 Results

3.1 Global distribution of SOC MRT

3.2 Model performance

3.3 Variable importance

3.4 Marginal effects on MRT

3.5 ESM benchmarking

4 Discussion

5 Conclusions

Data availability

Soil profile data are from WoSIS (https://www.isric.org/explore/wosis). ERA5-Land data are available from the Copernicus Climate Data Store (https://cds.climate.copernicus.eu). SoilGrids data are available from ISRIC (https://soilgrids.org). ESA WorldCover is available at https://worldcover2021.esa.int. Hansen forest change data are available at https://earthenginepartners.appspot.com/science-2013-global-forest. MODIS products are available from the LP DAAC. Microbial datasets are available from their respective repositories as cited in the text.

Code availability

The complete R analysis pipeline (18 scripts) is archived at [DOI/repository].

Acknowledgements

Author contributions

Competing interests

The authors declare no competing interests.

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Figures

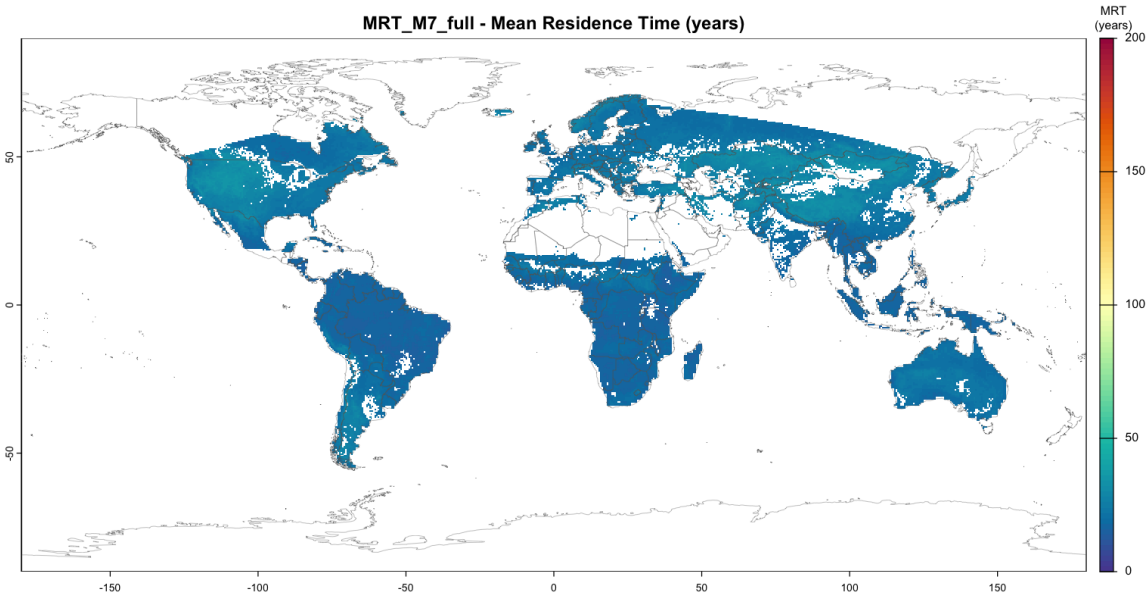

Figure 1: **Global distribution of predicted soil organic carbon mean residence time (MRT).**

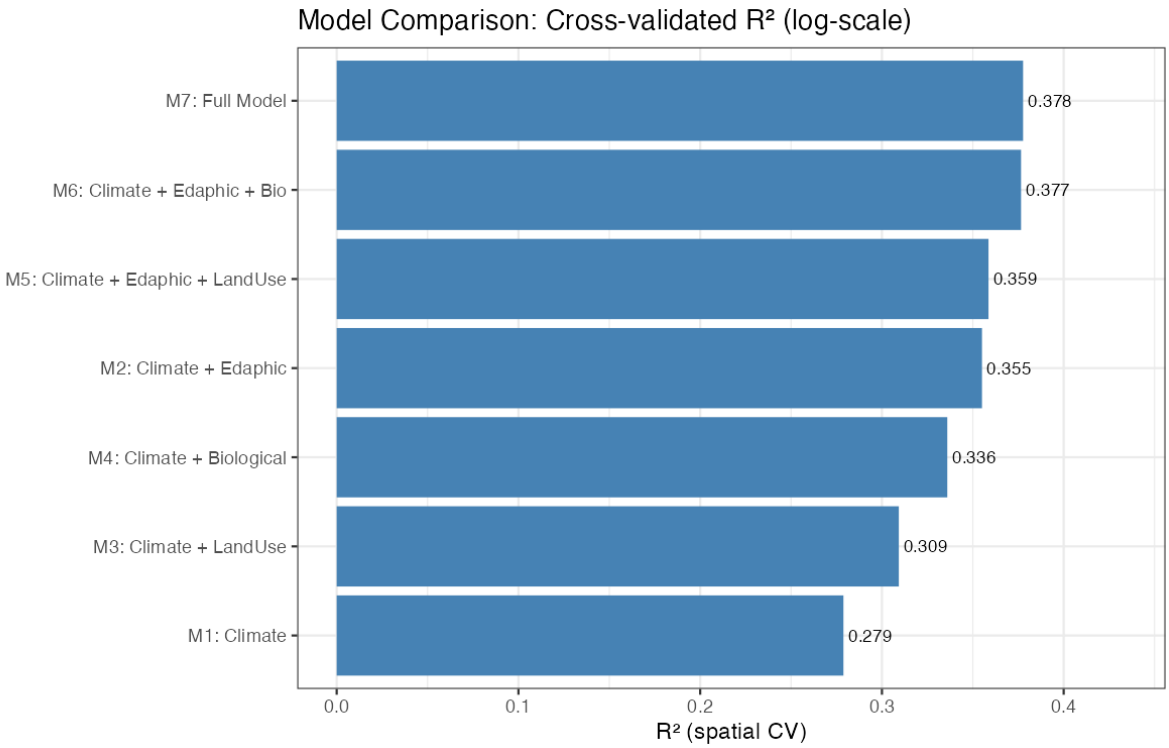

Figure 2: **Cross-validated model performance across nested configurations (M1–M7).**

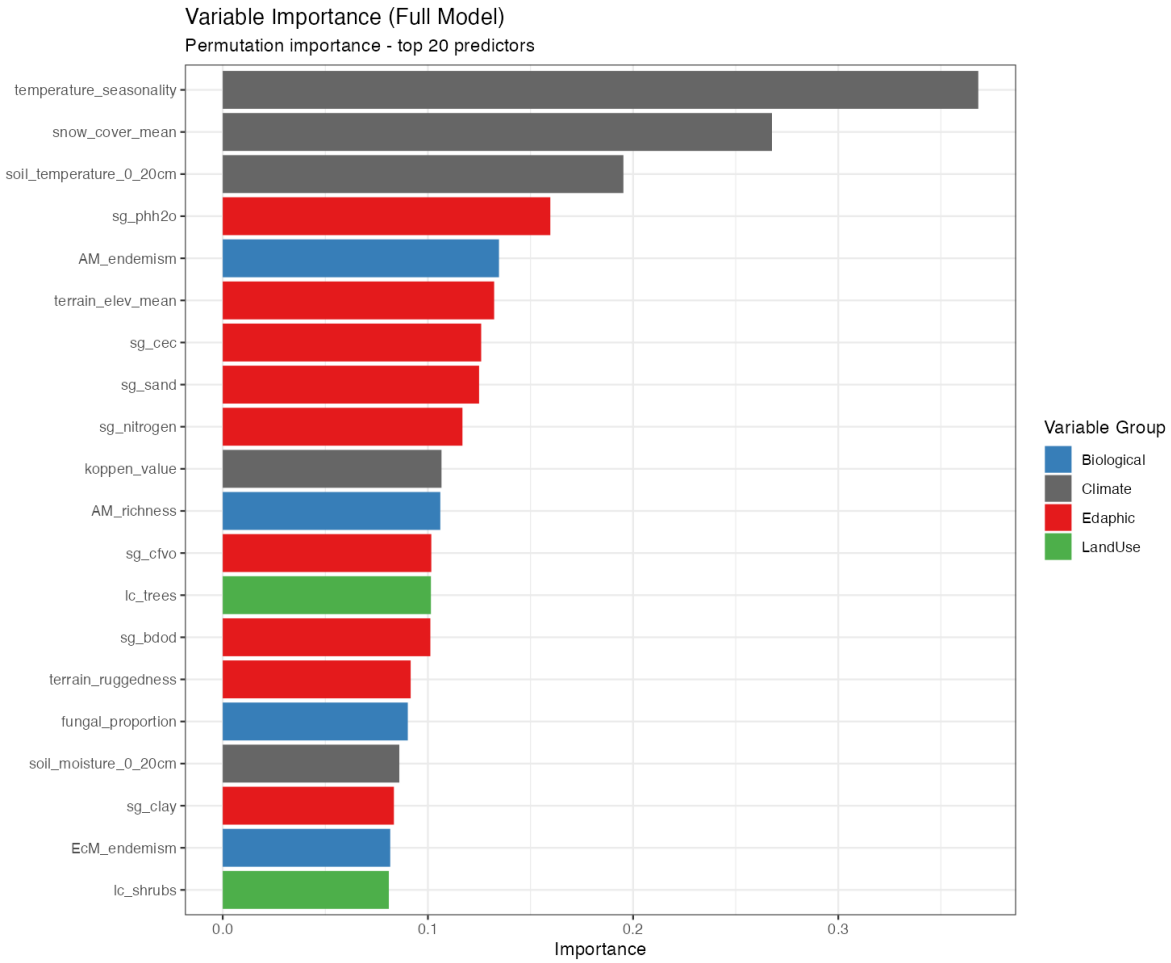

Figure 3: **Permutation variable importance for the full model (M7).**

Tables

Table 1: **Model configurations and cross-validated performance metrics.** All models were fitted using `ranger` (500 trees, min. node size 5) with log-transformed MRT as the response variable. Spatial block cross-validation used 2° grid cells assigned to ten folds. ΔR^2 is relative to M1 (Climate only). Performance metrics are on the log scale.

Model	Variable groups	n_{pred}	R^2_{CV}	RMSE_{CV}	R^2_{OOB}	ΔR^2
M1	Climate	6				—
M2	Climate + Edaphic	19				
M3	Climate + Land Use	14				
M4	Climate + Biological	15				
M5	Climate + Edaphic + Land Use	27				
M6	Climate + Edaphic + Biological	28				
M7	Full model	36				

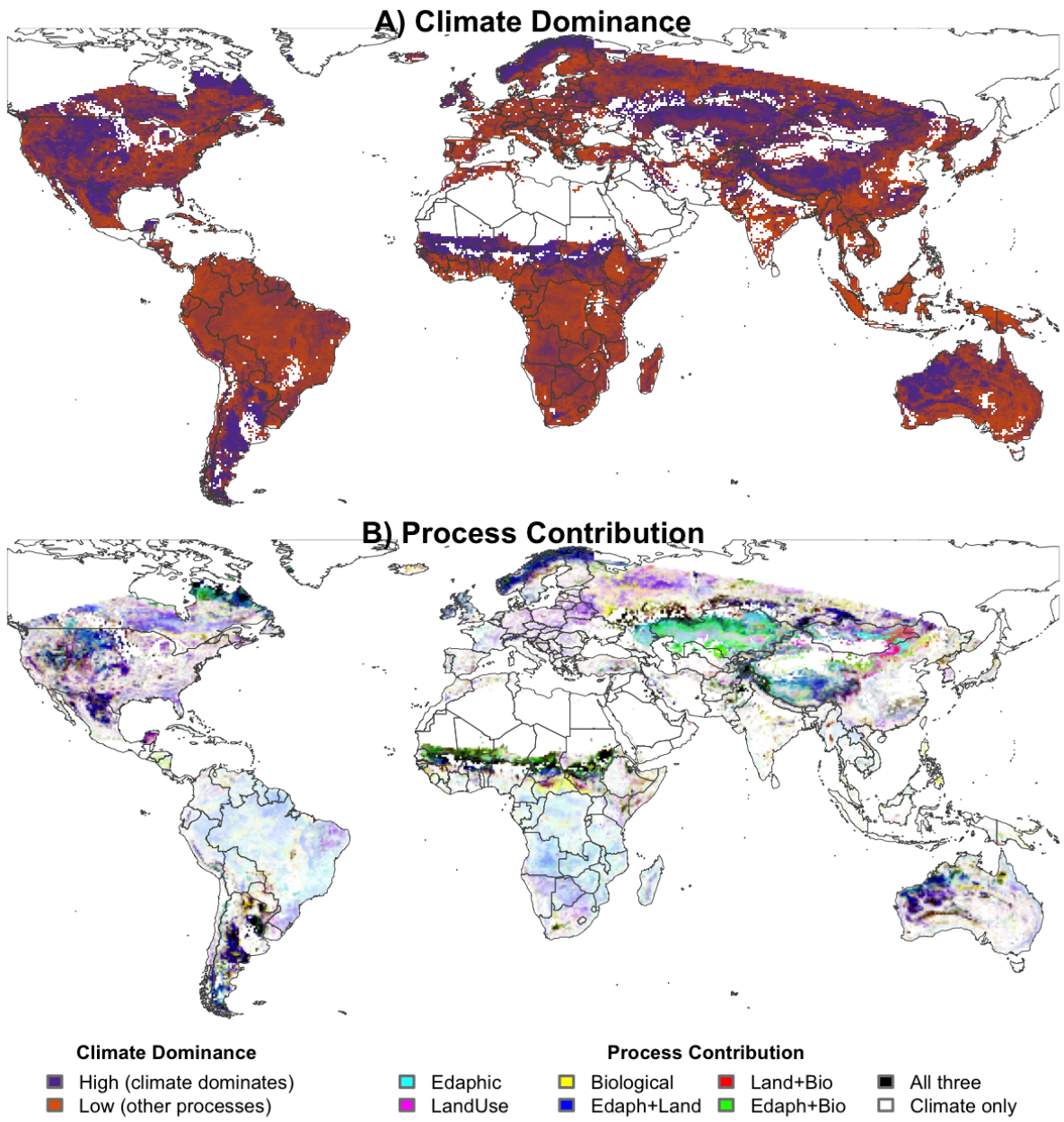

Figure 4: **Spatial dominance of process classes driving SOC mean residence time beyond the climate baseline.**

A Supplementary figures

B Supplementary tables

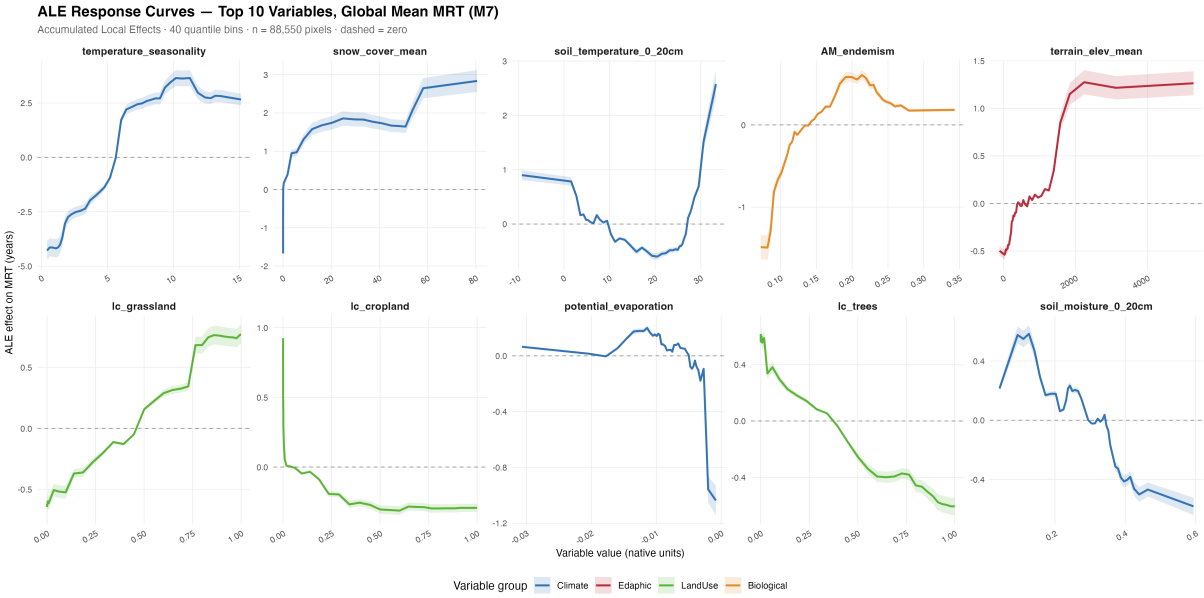

Figure 5: **Accumulated Local Effects (ALE) response curves for the ten most responsive predictors of SOC mean residence time.**

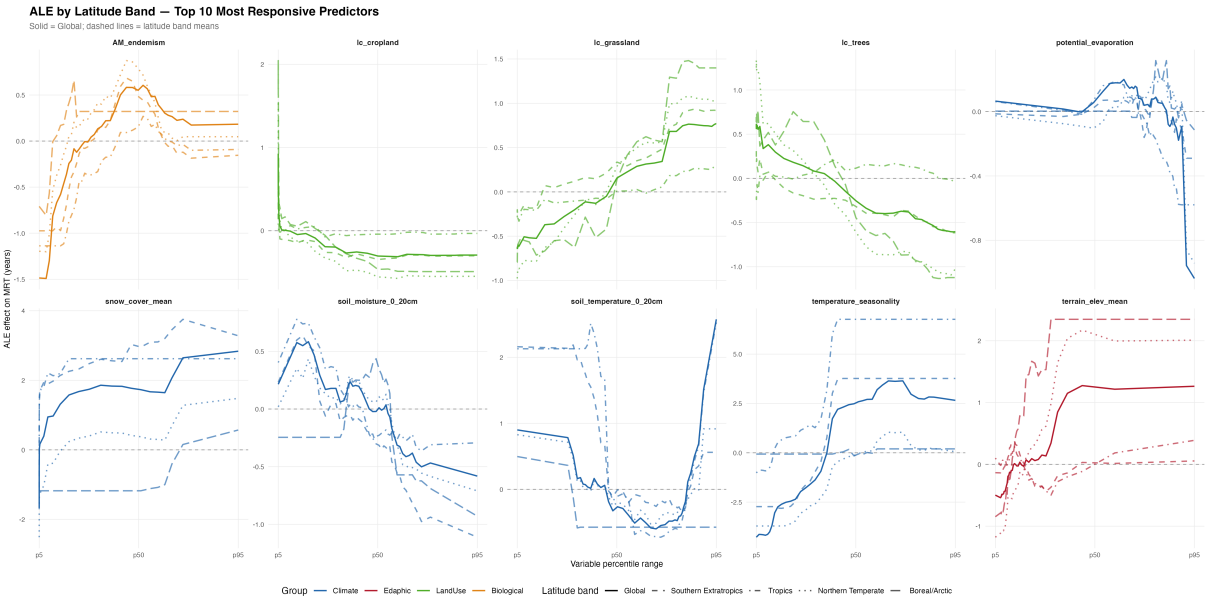

Figure 6: **Latitude-band stratification of ALE response curves for the ten most responsive predictors.**

Table 2: **Predictor variables in the full model (M7), organised by mechanistic group.** Native resolution refers to the original data product; all variables were extracted at native resolution where possible and resampled to 0.1° for global prediction.

Group	Variable	Units	Source
Climate	Soil temperature (0–20 cm)	°C	ERA5-Land (depth-weighted stl1/stl2)
	Temperature seasonality	K	ERA5-Land (monthly SD of t2m)
	Soil moisture (0–20 cm)	$m^3\,m^{-3}$	ERA5-Land (depth-weighted swvl1/swvl2)
	Snow cover fraction	0–1	ERA5-Land
	Potential evaporation	m	ERA5-Land
	Köppen-Geiger zone	1–30	Beck et al. (2023), 1991–2020
Edaphic	Clay content	%	SoilGrids 250 m v2.0, 0–20 cm
	Sand content	%	SoilGrids 250 m v2.0, 0–20 cm
	Bulk density	$kg\,dm^{-3}$	SoilGrids 250 m v2.0, 0–20 cm
	Coarse fragment volume	%	SoilGrids 250 m v2.0, 0–20 cm
	pH (H_2O)	—	SoilGrids 250 m v2.0, 0–20 cm
	Cation exchange capacity	$cmol\,kg^{-1}$	SoilGrids 250 m v2.0, 0–20 cm
	Total nitrogen	$g\,kg^{-1}$	SoilGrids 250 m v2.0, 0–20 cm
	WRB functional group	1–8	SoilGrids WRB, reclassified
	Elevation	m	Copernicus GLO-30 DEM
	Slope	°	Copernicus GLO-30 DEM
	Northness	−1 to +1	Copernicus GLO-30 DEM
	Eastness	−1 to +1	Copernicus GLO-30 DEM
	Terrain ruggedness	m	Copernicus GLO-30 DEM (focal SD)
Land Use	Tree cover fraction	0–1	ESA WorldCover 2021
	Grassland fraction	0–1	ESA WorldCover 2021
	Shrubland fraction	0–1	ESA WorldCover 2021
	Cropland fraction	0–1	ESA WorldCover 2021
	Bare/sparse fraction	0–1	ESA WorldCover 2021
	Wetland fraction	0–1	ESA WorldCover 2021
	Forest loss (binary)	0/1	Hansen GFC v1.10
	Fire count (pre-sampling)	count	MODIS MCD64A1, 2001–2020
Biological	Fungal proportion	0–1	Yu et al. (2022)
	AM root colonisation	%	Barceló et al. (2023)
	EcM root colonisation	%	Barceló et al. (2023)
	EcM:AM root ratio	—	Derived from above
	AM fungal richness	index	Van Nuland et al. (2025) (SPUN)
	EcM fungal richness	index	Van Nuland et al. (2025) (SPUN)
	AM endemism (RWR)	index	Van Nuland et al. (2025) (SPUN)
	EcM endemism (RWR)	index	Van Nuland et al. (2025) (SPUN)
	EcM:AM richness ratio	—	Derived from above

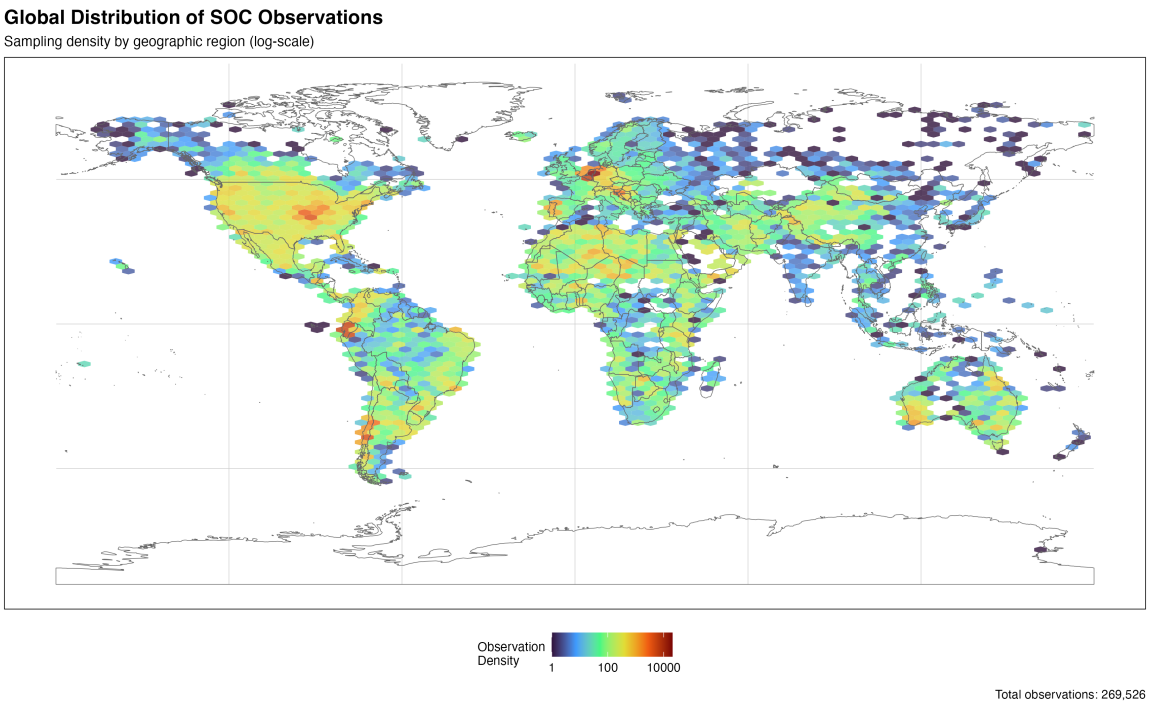

Figure 7: **Global distribution and density of soil organic carbon observations used for model training.**

Table 3: **Belowground NPP allocation fractions (f_{BNPP}) by dominant land cover class used to estimate soil carbon input.** Values represent the fraction of total NPP entering the soil as belowground carbon (roots, exudates, mycorrhizal transfer).

Land cover class	ESA code	f_{BNPP}
Tree cover	10	0.35
Shrubland	20	0.45
Grassland	30	0.55
Cropland	40	0.30
Built-up	50	0.20
Bare/sparse	60	0.30
Herbaceous wetland	90	0.50
Mangroves	95	0.40
Moss/lichen	100	0.60
Default (missing)	—	0.40